Programming Fundamentals

Lab 1 Submission

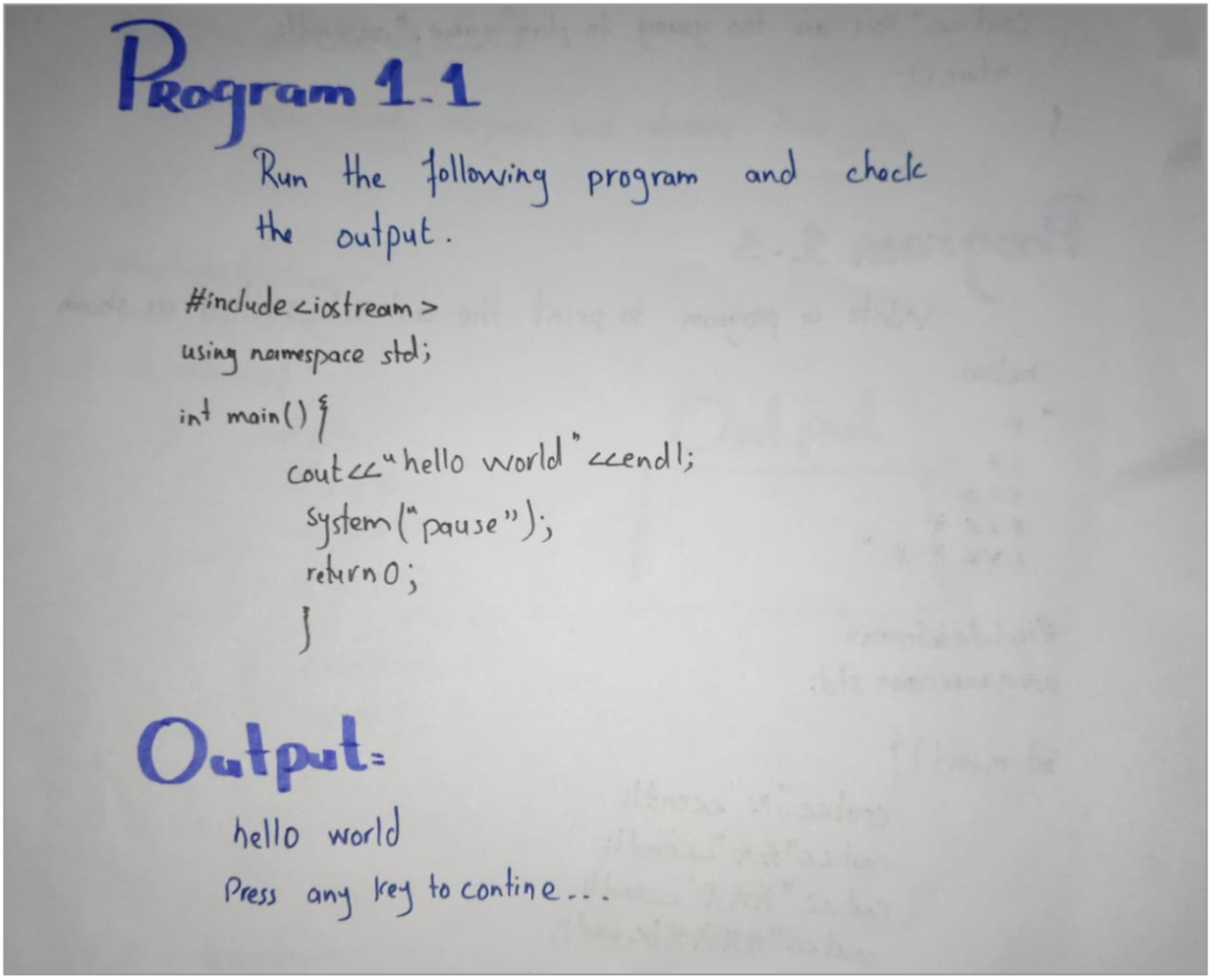
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Date	9 October 2026

Program 1.1

Question

Run the following program and check the output.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "hello world" << endl;
    system("pause");
    return 0;
}
```

Output

```
hello world
Press any key to continue . . .
```

Explanation

- `#include <iostream>` adds the input/output library, which gives us `cout` for printing.
- `using namespace std;` lets us write `cout` and `endl` directly instead of `std::cout`.
- `int main()` is the main function. Program execution always starts here.
- `cout << "hello world" << endl;` prints hello world on the screen, and `endl` moves the cursor to the next line.
- `system("pause");` holds the console window open and shows "Press any key to continue . . ." so we can read the output.
- `return 0;` ends the program and tells the system it finished successfully.

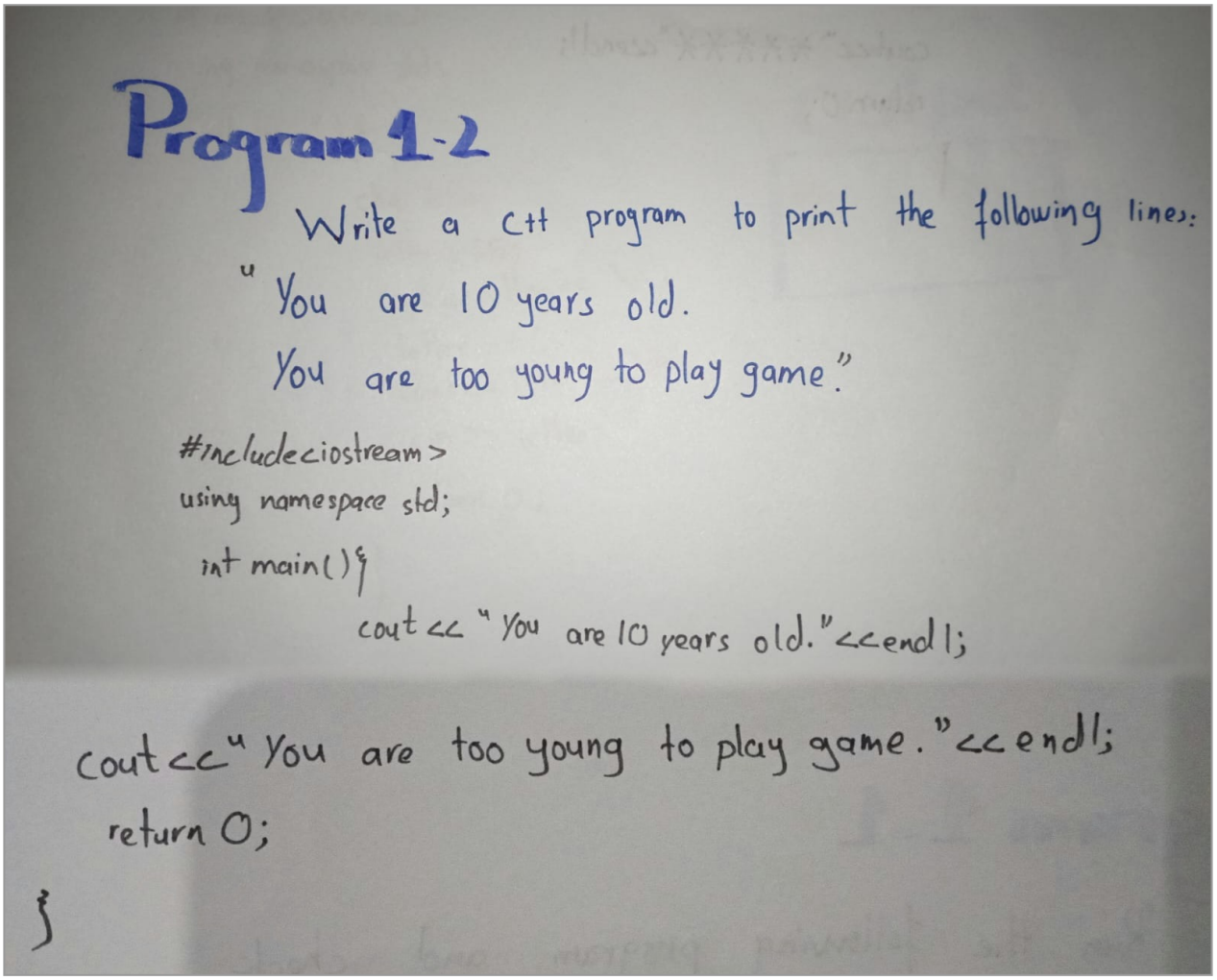
Program 1.2

Question

Write a C++ program to print the following lines:

```
You are 10 years old.
You are too young to play game.
```

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "You are 10 years old." << endl;
    cout << "You are too young to play game." << endl;
    return 0;
}
```

Output

```
You are 10 years old.
You are too young to play game.
```

Explanation

- The program uses two `cout` statements, one for each line we want to print.
- Text written inside double quotes is printed exactly as it is written.
- `endl` at the end of the first line moves the cursor to a new line, so the second sentence appears below the first one. Without `endl`, both sentences would print on the same line.
- `return 0;` ends the program after both lines are printed.

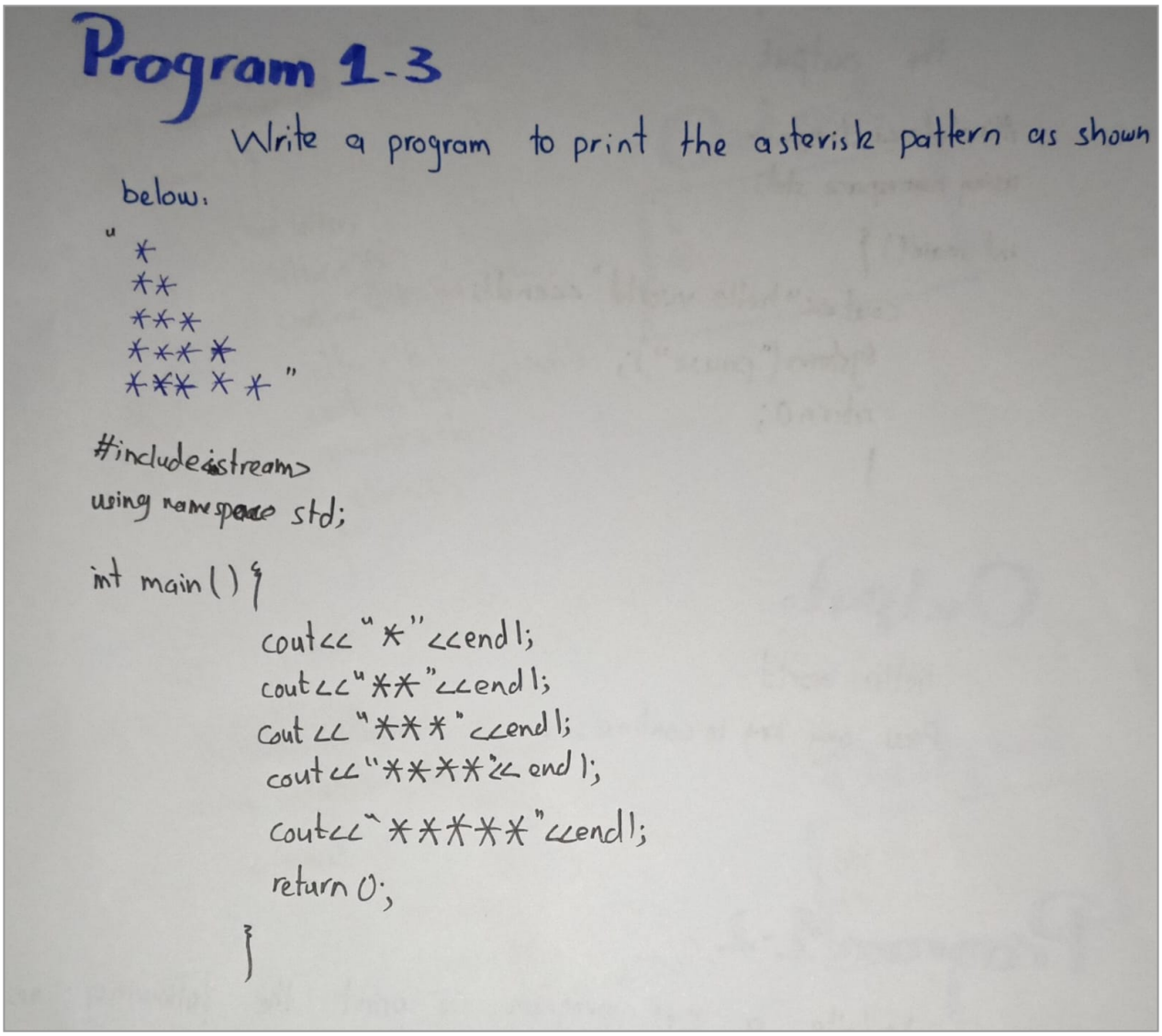
Program 1.3

Question

Write a program to print the asterisk pattern as shown below:

```
*
**
***
****
*****
```

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "*" << endl;
    cout << "**" << endl;
    cout << "***" << endl;
    cout << "****" << endl;
    cout << "*****" << endl;
    return 0;
}
```

Output

```
*
**
***
****
*****
```

Explanation

- Each `cout` statement prints one row of the pattern.
- The number of asterisks increases by one in every row (1, 2, 3, 4, 5), which makes the triangle shape.
- `endl` after each row starts the next row on a new line.
- Here every row is written by hand. The same pattern can also be printed with nested `for` loops, which is useful when the pattern has many rows.

END OF ASSIGNMENT

```cpp
return 0;   // program finished successfully
```

Thank you for reading!